HYBRID DATA ARCHIVAL STRATEGY: A COMPARATIVE STUDY

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1. AIM

The aim of the study is to perform a comparative analysis of the proposed Hybrid Archival Strategy and a Time-Based Archival Strategy using a synthetically generated sensor dataset.

2. OBJECTIVE

- To implement a conventional Time-Based Archival Strategy based on the number of days elapsed since the last access of a record.
- To propose a comprehensive Hybrid Archival Strategy integrating the storage utilisations, number of days since the last access and the access frequency over the preceding 100 days to arrive at the archival decision.
- To import a synthetically generated dataset into the Oracle Database Environment to the comparative study.
- To evaluate and compare the two strategies against relevant performance metrics.
- To determine whether the proposed multi-factor hybrid model provides a measurable enhancement over the conventional Time-Based Strategy to the database management system.

3. ABSTRACT

With the rapid expansion of data volume in database systems, monitoring data growth and implementing appropriate mechanisms to maintain the efficiency of the database systems has become a non-negotiable part of Database Management. The continuous generation and lack of proper management of data causes infrequently accessed data to get accumulated over time. Suitable data management and organisation is essential for optimal data to value generation and optimal utilisation of available resources. Effective storage management can include multi-temperature data management where data is handled as per its access frequency. Frequently accessed data can be stored on active, fast storage devices while data that is not actively accessed for operations in a given timeframe can undergo systematic archival to reduce the storage burden on the database. Data archival, hence forms an important component of the Database Lifecycle Management, particularly for systems that generate large volumes of data in small timeframes. Conventional time-based archival strategies are restricted to temporal characteristics of data, like the number of days since the record was last accessed. Though, the age of data provides an essential indication of its inactivity, this parameter alone is insufficient in distinguishing cold data from actively accessed data and deciding its archival eligibility. Records with similar age may have significantly different access patterns and storage

requirements. An actively required record may inappropriately be archived solely on the basis of one factor, increasing the overhead of retrieving it from the slower storage device when it is required. A time-based strategy does not consider whether a record occupies a relatively higher storage and therefore, could contribute significantly to storage optimisation if appropriately identified for archival. These limitations motivate the investigation of a multifactorial archival mechanism. Hence, we propose a comprehensive Hybrid Archival Strategy that takes into account three characteristics of a record: the Database Storage Utilisation by the record, Number of days since the record was last accessed and the Access Frequency over the preceding 100 days. A Hybrid Archival Score is calculated by integrating the factors into a proposed mathematical formula. The mathematical model is proposed considering that storage utilisation and inactivity of data contribute positively towards archival eligibility, while recent access frequency has an inverse influence on it. A threshold is consequently established to decide whether the record should be archived or kept in the main database. The conventional time-based strategy is implemented in parallel for a comparative analysis of the two mechanisms. The strategies are applied on a synthetically generated sensor dataset, the simulation results of which, show that the proposed hybrid archival strategy can improve the cold capture rate by 6.9063% in a system of 100,000 sensor readings. Relevant attributed are derived from the available variables to calculate the days since last access, access frequency and storage utilisation for each record. The evaluation focuses on the number of actively accessed records archived and the cold capture rate of the strategies to measure their effectiveness and accuracy in identifying records for archival. The study demonstrates the feasibility of multi-factor archival decision model using SQL operations. It also establishes the foundation for further investigation into adaptive archival mechanisms to match the requirements of any given database.

4. INTRODUCTION

The explosive growth in data volume across the globe has put an enormous pressure on the need to inculcate more efficient data management strategies. Under these circumstances, database systems must continue to provide efficient execution of queries and updates to the database while maintaining the read and write performance of the system. The database system must provide specialised functionalities and data structures to speed up desired manipulation to the data. Relevant data management strategies are particularly essential in systems that continuously generate data, such as sensor-based applications, where data may get accumulated over time.

Data Lifecycle Management helps monitor data as it passes through a series of stages from its creation to deletion. It helps manage data in each phase and helps establish appropriate mechanisms to maintain data in optimal conditions. Data archival and retention is a pivotal section of the data lifecycle. Depending on the user's access patterns, certain records become irrelevant for everyday operations over time. Such data hinders insertion, deletion or modification in the database and consumes storage that can otherwise be used by other relevant records. Organisations maintain copies of such data in a separate long-term storage repository

called an archive. Archived data can be retrieved and restored to the central, active database when required. However, defining what data should be sent to archive is a crucial segment of the data archival process. Treating every record equivalently is an inefficient way of managing a database. Hence, suitable strategies must be introduced for ideal data archival.

The conventional Time-Based Archival Strategy considers only the number of days since the last access to conclude whether it is best to send a record for archival. An appropriate data management technique is expected to align with a diverse array of factors that influence the performance of the database system. Insufficient representation of such factors in the decision making can lead to inappropriate archival judgements. It is important that the mechanism rightly segregates cold data from actively accessed data. Presence of active data in the archive will conversely increase the overhead of retrieving it from the archive and defeat the purpose of the model.

Hence, we propose a comprehensive Hybrid Archival Strategy that inculcates Storage Utilisation by the record, Number of days since the last access and the Access Frequency of the record over the preceding 100 days to arrive at the archival decision. The proposed strategy is implemented within an Oracle Database environment on a synthetically generated sensor dataset. The dataset includes sensor readings along with access log information, allowing the inclusion of both storage and data access related characteristics.

The proposed Hybrid Archival Strategy is evaluated against the Time-Based Strategy to compare their efficiencies. The analysis focuses on the accuracy of archival decision over the quantity of records archived. The percentage of cold data sent to the archive and the retention of active data in the database have been applied as the metrics to quantify the efficiencies of the two approaches. The study emphasizes not only on implementing the Hybrid Archival Model as the better option, but also establishing the parameters used as influential figures in database management.

5. RELATED MATERIAL / LITERATURE REVIEW

The ever-increasing global volume of data has made it crucial for the database administration to implement efficient data storage and management mechanisms. Yan and Wang [8] examined the archival of historical sensor data highlighting the challenges associated with the growth in data due its continuous generation and accumulation in a short timeframe. Their study presents memory allocation, data compression and the designing of efficient storage structures as essential segments of the archival mechanism, helping understand the essence of data archival.

Ou [9] examined data storage management during the data lifecycle. The study highlighted the importance of temporal factors to implement suitable storage management decisions. This serves as the basis for archival strategies aligned with time where the degradation of data value with time is taken into consideration.

Cheng et al. [11] implemented and performed a performance analysis of an archival system for the Taiwan Photon Source Project. The study explains the significance of archival systems,

especially in databases generating large volumes of data. It also helps understand the mechanisms and parameters for the quantitative analysis of archival strategies.

Vasile et al. [12] evaluated modern time-series databases for archiving data associated with ATLAS experiment. The study presents the importance of relevant database structures and technologies to handle large scale data. It helps establish the relevance and importance of appropriate archival strategies.

Song et al. [10] proposed a Hybrid Data storage method based on the data temperature and global disk-space utilisation. The study is significantly relevant to our study as it integrates different characteristics to arrive at an enhanced mechanism. It provides conceptual support to our proposal and analysis as it discusses the importance of combining different parameters.

Sinaeepourfard et al. [17] proposed a comprehensive model for data lifecycle, specialised for Big Data environment. Their work creates emphasis on the different phases that the data passes through from creation to deletion and the appropriate mechanisms required for its management in each phase. Data archival being an important component of the data lifecycle, makes the study relevant to our proposal and analysis. It also helps gauge the influence of time through the data lifecycle.

Arass et al. [16] analysed different data lifecycle models and how data management demands vary as per the characteristics and structure of data. This supports the claim to inculcate flexible mechanisms rather than staying rigid to a single parameter.

Aghaei et al. [18] presented the best practices for managing passive acoustics data at Ocean Networks Canada. The study helps design and implement relevant data management mechanisms for inactive data records that have accumulated over time.

Nunome and Hirata [19] analysed the migration efficiency in distributed storage systems using dynamic tiering. The study investigates the overhead of the movement of data between storage levels. This helps understand the significance of avoiding unnecessary archival and increasing the overhead of retrieving data.

These studies demonstrate that effective archival requires the consideration of other data characteristics that are vital to the data lifecycle. Limited emphasis has been placed on combining storage utilisation, access frequency and number of days since last access for data archival, which the forms the research gap addressed in the proposed Hybrid Archival Strategy.

6. PROPOSED METHODOLOGY

The conventional Time-Based Archival Strategy and the proposed Hybrid Archival Strategy were implemented on a synthetically generated sensor dataset containing sensor readings along with their access log information using SQL queries in Oracle Database Environment. The comparative analysis of the two strategies was carried out against relevant parameters to conclude whether a multi-factor archival model can outperform a unifactorial model.

6.1. SYSTEM ARCHITECTURE

A sensor dataset of 100,000 readings and 3,500,000 access events was synthetically generated using the Pandas and Numpy libraries of Python, providing the core foundation of the study. The distribution of access patterns of data was manipulated for sufficient representation and performance analysis of the strategies under different conditions.

Each sensor reading is supported by relevant attributes like Reading ID and size of the data. A separate Data Access Log records the timestamp of access of each reading. Fig. 6.1.1. depicts the Entity Relationship Model of the given schema.

The datasets help retrieve and derive appropriate variables for the analysis. The Time-Based Archival Strategy uses the number of days since last access to judge the eligibility of a record for archival while the proposed Hybrid Data Archival Strategy arrives at an archival decision aligned with the storage utilisation by the record, data access frequency in the last 100 days, and the number of days since last access.

The performance of the two mechanisms is monitored using predefined evaluation metrics.

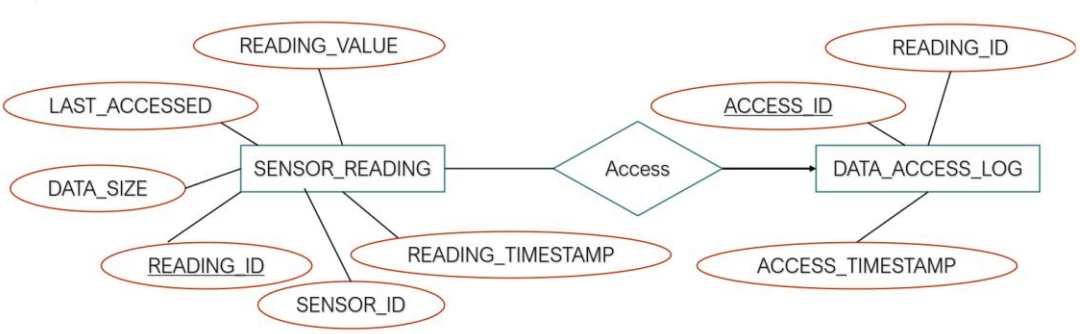


Figure 6.1.1- Entity-Relationship Model of the dataset

6.2. MATHEMATICAL MODEL

6.2.1. Time-Based Archival Model

For each sensor reading i, the number of days since last access, D_i is given by:

D_i = Current Date - $Last_Accessed_i$

A threshold of 230 days was established for the given model.

$D_i \geq 230$ (Send for Archival)
$D_i < 230$ (Keep in database)

Thus, the archival decision is solely based on the age of the record since last access.

6.2.2. Hybrid Archival Model

The hybrid archival model takes into consideration three factors for the decision making. A Hybrid archival score, H_i has been implemented to obtain a quantitative measure of the influence of each factor on the archival decision.

1. Storage Utilisation by the record, U_i

$U_i = \frac{Record\ size}{Database\ Capacity}$

This parameter represents the fraction of database capacity that the record will occupy if retained in the database.

2. Number of days since last access, D_i

D_i = Current Date - Last_Accessed$_i$

This parameter represents the number of days elapsed since the data was last accessed.

3. Access frequency in the last 100 days, f_{100_i}

$f_{100_i} = \frac{A_{100_i}}{100}$,

where A_{100_i} is the number of accesses over the preceding 100 days.
This parameter helps understand the recent user access pattern surrounding the record.

Hybrid Archival Score, H_i

The three factors are integrated into a mathematical formula to calculate the Hybrid Archival Score as follows:

$H_i = \frac{U_i \ln(D_i+1)}{f_{100_i}+0.01}$

U_i and D_i have a positive correlation while f_{100_i} has an inverse correlation with the Hybrid Archival Score. A higher number of days since the last access and a lower access frequency over the preceding days point to the inactivity of the record in the given timeframe, increasing its eligibility for archival.

Records with comparable access behaviour can be compared by the fraction of storage they utilise in the database. A record with a higher storage utilisation is given a leverage over the others for the archival as it is consuming a larger proportion of space in the database.

6.3. COMPARATIVE EVALUATION

Both the strategies were assessed over the same dataset. The corresponding thresholds were established such that the models send a comparable amount of data for archival. The objective of the analysis is not to determine the model sending a larger quantity of data for archival, but the model sending an improved quality of data for archival. Sending actively accessed data for archival increases the overhead of retrieving it and jeopardises the purpose of the mechanism.

Hence, we compare the capacities of the models to distinguish cold data from actively accessed data and the retention of active data in the database.

The following evaluation metrics were considered:

1. Number of records archived
2. Number of cold records archived
3. Number of active records archived

4. Cold Data Capture Rate
5. Active-data retention/ reduction in active records sent for archival

7. RESULT & DISCUSSION

Table 7.1. depicts the archival decisions by the Time-Based and Hybrid Strategies

READING_ID	DATA_SIZE	DAYS_SINCE_ACCESS	ACCESS_COUNT_100D	TIME_BASED_DECISION	HYBRID_DECISION
59585	114	183	1	KEEP	KEEP
59147	96	532	1	ARCHIVE	KEEP
55583	164	285	0	ARCHIVE	ARCHIVE
47880	191	193	5	KEEP	KEEP
86775	82	89	1	KEEP	KEEP
49718	159	57	27	KEEP	KEEP
29302	138	70	45	KEEP	KEEP
5920	139	19	90	KEEP	KEEP
71037	83	460	0	ARCHIVE	KEEP
30471	134	18	162	KEEP	KEEP

Table 7.1-Sample Records showing archival decisions by the two strategies

7.1. COMPARISON OF ARCHIVAL DECISIONS

The archival strategies were first compared on the basis of records with different archival decisions by the two mechanisms. The records where one strategy concluded that the record should be sent for archival and the other concluded that the record should be kept in the main database were extracted to analyse the differences in the decision-making mechanisms.

Fig. 7.1.1 depicts a snippet of records with contradicting archival decisions from the two mechanisms.

	READING_ID	DATA_SIZE	DAYS_SINCE_ACCESS	ACCESS_COUNT_100D	ACCESS_FREQUENCY	U_I	H_SCORE	TIME_BASED_DECISION	HYBRID_DECISION
1	33097	200	228	0	0	0.009766	5.3064	KEEP	ARCHIVE
2	36059	200	224	0	0	0.009766	5.2892	KEEP	ARCHIVE
3	85255	199	228	0	0	0.009717	5.2798	KEEP	ARCHIVE
4	83601	199	226	0	0	0.009717	5.2713	KEEP	ARCHIVE
5	21165	199	224	0	0	0.009717	5.2627	KEEP	ARCHIVE
6	53529	199	222	0	0	0.009717	5.254	KEEP	ARCHIVE
7	26017	198	227	0	0	0.009668	5.2491	KEEP	ARCHIVE
8	15566	198	226	0	0	0.009668	5.2448	KEEP	ARCHIVE
9	6639	198	226	0	0	0.009668	5.2448	KEEP	ARCHIVE
10	66081	198	226	0	0	0.009668	5.2448	KEEP	ARCHIVE

Figure 7.1.1- SQL Output snippet of the disagreements between the two strategies

Fig. 7.1.2 presents a graphical representation of the disagreement between the two strategies.

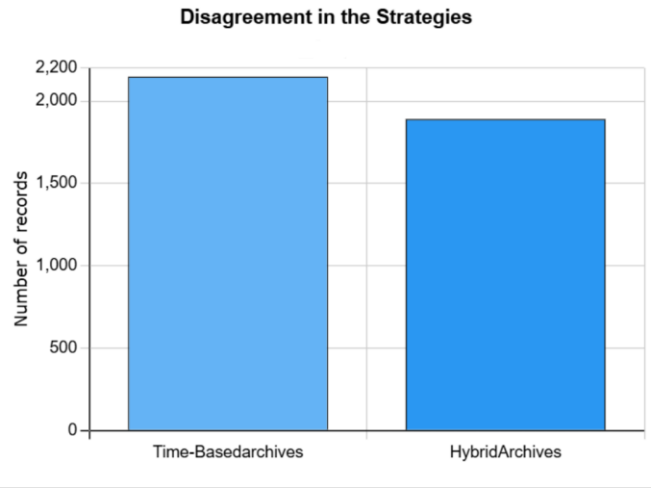


Figure 7.1.2- Graphical representation of the difference in archival decisions by the two strategies

7.2. QUANTITATIVE COMPARISON

The two strategies were then compared using the predefined evaluation parameters. Attributes taken into consideration are the number of records archived, number of cold records archived, number of active records archived, and the efficiency of capturing cold records for archival.

Fig. 7.2.1 presents the SQL output of the quantitative comparison of these metrics

	STRATEGY	TOTAL_ARCHIVED	COLD_ARCHIVED	ACTIVE_ARCHIVED	COLD_CAPTURE_RATE
1	TIME-BASED	18352	16267	2085	88.32
2	HYBRID	18095	17539	556	95.23

Figure 7.2.1- SQL Output Snippet of the quantitative comparison

7.3. COLD AND ACTIVE DATA ARCHIVAL

The distribution of cold and active records among the archived records provides a clear indication of the efficiency and accuracy of the archival decisions. Cold records are defined as those with no accesses during the preceding 100 days, whereas active records are defined to have at least one access during the same timeframe.

Fig. 7.3.1 provides a visual comparison of the archival based on the parameters.

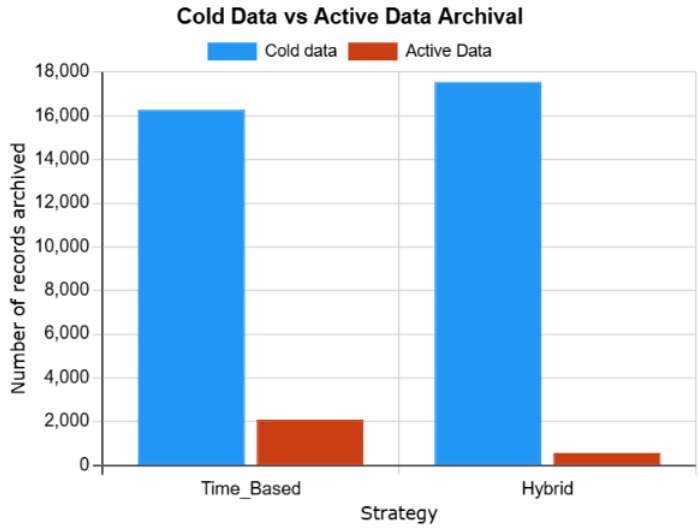


Figure 7.3.1- Graphical Comparison of Archival of Cold Data and Active data by the two strategies

The results indicate that although the Time-Based Strategy archives a slightly larger number of records than the Hybrid Archival Strategy, a greater proportion of its archived records consist of actively accessed records. In contrast, the Hybrid Archival Strategy more actively sends cold records for archival while reducing the number of active records unsuitably sent for archival.

7.4. COLD DATA CAPTURE RATE

The Cold Data Capture Rate was used to evaluate the effectiveness of the strategies in identifying cold records based on their recent access patterns.

Fig. 7.4.1 illustrates a graphical representation of the percentage of cold data archived by the two strategies.

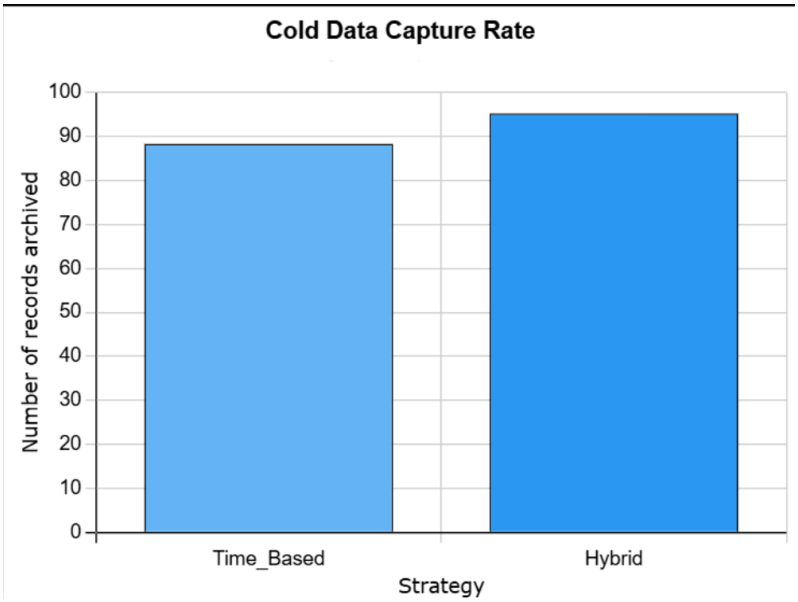


Figure 7.4.1- Graphical Representation of the percentage of cold data archived by the two strategies

The Hyrid Archival Strategy achieved a higher accuracy in capturing cold data for archival as compared to the Time-Based Strategy. This indicated the positive influence of integration of

other factors along with the time since last access in the Hybrid Archival Strategy and how is helps achieve better archival results.

7.5. DISCUSSION

The results help establish that the number of records sent for archival is not a sufficient factor to judge the efficiency of an archival model. While the Time-Based Strategy showed a slightly higher archival rate than the Hybrid Archival Strategy, it also sends a larger number of actively accessed data to the archive.

The Hybrid Archival Strategy, on the other hand, considers multiple characteristics of each record, including storage utilisation, last access, and access frequency to reach the archival decision. This enables it to weigh and integrate relevant parameters to decide whether a record is cold or active.

Therefore, the comparative analysis suggests that the proposed, comprehensive multi-factor approach provides a more selective and quality-focused archival mechanism over the conventional Time-Based Strategy

8. CONCLUSION

The study presented a comparative analysis of the conventional Time-Based Archival Strategy and the proposed Hybrid Archival Strategy over a synthetically generated sensor reading dataset in an Oracle Database environment. While the Time-Based Strategy arrives at an archival decision solely on the basis of the number of days elapsed since the last access, the comprehensive Hybrid Archival Strategy integrates the Storage Utilisation, Access frequency over the preceding 100 days and the Number of Days since the last access to make the archival decision. Both the strategies were evaluated on the same dataset to ensure uniformity and avoid bias. While the Time-Based Archival Strategy archives a slightly higher number of records than the Hybrid Strategy, it unsuitably sends 2.75 % more active records than the Hybrid Model for archival. Additionally, while the percentage of cold data sent for archival is 88.32% for the Time-based Model, it was observed to be 95.23 % for Hybrid Model. The archival of active records by the Hybrid Model was also 73.33% lower than that of the conventional Time-based Model. The Hybrid Archival Strategy having a higher Active data retention rate and Cold data archival rate demonstrates its feasibility over the Time-Based Model. The study henceforth also presents the significance of considering multiple factors in Archival Management. However, the data archival was only simulated in the study and the strategies can be further investigated in real-world archival conditions using actual storage systems. Overall, the study presents a promising foundation to an enhanced hybrid archival strategy.

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